

Linux / Bash — Consolidated Cheatsheet

1) Quick tips & shell basics

Command	Description	Example
man cmd	Read command manual	man grep
cmd --help / help cmd	Quick help (builtins use help)	grep --help / help cd
cmd1 && cmd2	Run second only if first succeeds	mkdir dir && cd dir
cmd1 cmd2	Run second if first fails	grep foo file echo 'not found'
cmd1 cmd2	Pipe output	ps aux grep nginx
\$(...)	Command substitution	files=\$(ls -l)
`...`	Legacy substitution	`date`
set -euo pipefail	Safer script defaults	set -euo pipefail
trap '...' EXIT	Cleanup on exit	trap 'rm -f \$TMP' EXIT
command -v prog	Check existence	command -v curl echo install

2) Filesystem navigation & basic file ops

Command	Description	Example
pwd	Print working directory	pwd
ls -lah	List files (human sizes, hidden)	ls -lah /var/log
cd /path	Change directory (cd - previous)	cd /etc && cd -
mkdir -p /path/to/dir	Create directory with parents	mkdir -p /srv/app/logs
rmdir dir	Remove empty directory	rmdir olddir
touch file	Create file or update mtime	touch /tmp/.heartbeat
cp -r src dest	Copy recursively; -p preserves attrs	cp -rp /srv/app /backup/app
mv src dest	Move/rename	mv old.conf new.conf
rm -rf dir	Remove dir (destructive)	rm -rf /tmp/mytest (■■ warning)
ln -s target linkname	Create symbolic link	ln -s /srv/app /opt/app
stat file	File metadata	stat /etc/hosts
file filename	Detect file type	file /usr/bin/python3

3) Viewing & editing text

Command	Description	Example
cat file	Print file	cat /etc/hosts
tac file	Print file in reverse	tac logfile
nl file	Number lines	nl script.sh
head -n 20 file	First 20 lines	head -n20 /var/log/syslog
tail -n 50 file / tail -f file	Last lines / follow	tail -f /var/log/nginx/access.log
less file	Pager with search	less /var/log/messages
sed 's/old/new/g' file	Stream edit (-i.bak for backup)	sed -i.bak 's/old/new/g' file
awk '{print \$1,\$3}' file	Field processing	awk -F: '{print \$1}' /etc/passwd
vim file / nano file	Editors	vim /etc/nginx/nginx.conf
printf '%s\n' "\$var"	Safe printing	printf '%s\n' "\$USER"

4) Searching (grep, find, locate, ripgrep)

Command	Description	Example
grep -Rin 'pattern' /path	Recursive grep with line numbers	grep -Rin 'ERROR' /var/log
egrep / grep -E	Extended regex	grep -E 'foo bar' file
fgrep / grep -F	Fixed-string search	grep -F 'literal' file
find /path -name '*.log'	Find files by name	find /var/log -name '*.log'
find ... -mtime +30 -exec gzip {} \;	Find & exec	find /var/log -type f -mtime +30 -exec gzip {} \;
find . -path './.git' -prune -o -type f -name '*.py' -print	Exclude '.git'	find . -path './.git' -prune -o -type f -name '*.py' -print
locate filename	Fast DB search (updatedb)	locate passwd
xargs	Build/run commands from stdin	find . -name '*.tmp' -print0 xargs -0 rm -f
rg pattern	ripgrep (fast)	rg 'TODO'